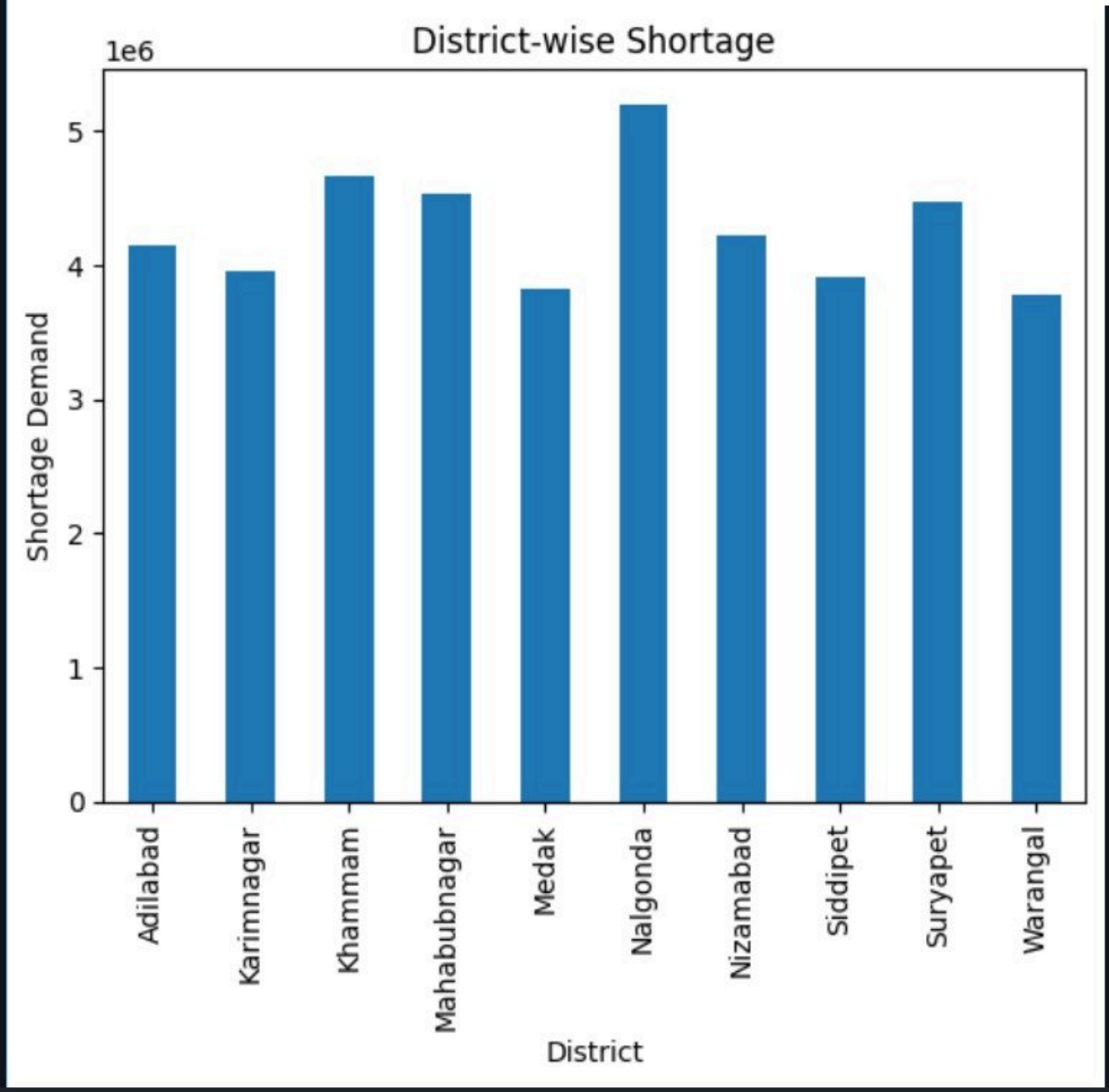
District


District-wise Shortage
1e6
5
4
3
2
1
0
Shortage Demand
Adilabad
Karimnagar
Khammam
Mahabubnagar
Medak
Nalgonda
Nizamabad
Siddipet
Suryapet
Warangal
District

Microsoft Azure
databricks
POS Data
Table
RationType
Allocated(Tonnes)
1 Rice 556
2 Wheat 456
District-wise Ration Demand & Shortage Risk
District Demand Trends (Next 30 days)
Adilabad
Karimnagar
Nizamabad
Warangal
Nalgonda
Hyderabad
Rangareddy
Khammam
Projected Demand over next 30 days
Shortage Risk Heatmap
Ration Type Allocation
Stock
Demand
Alert: Potential Shortage
Rice-NFSA
Wheat
RationType
Predicted Demand vs Available Stock
District Inventory Charts
Alert: Warangal stock levels critical
Warangal
Nalgonda
Potential Shortage
Predicted Monthly Demand
Current Available
District
Beneficiary Distribution Analysis
Card-type Distribution
Beneficiary Demographics
Age Distribution
Family Size
Spark jobs
Calculating district allocation profiles
Joining stock with demand tables

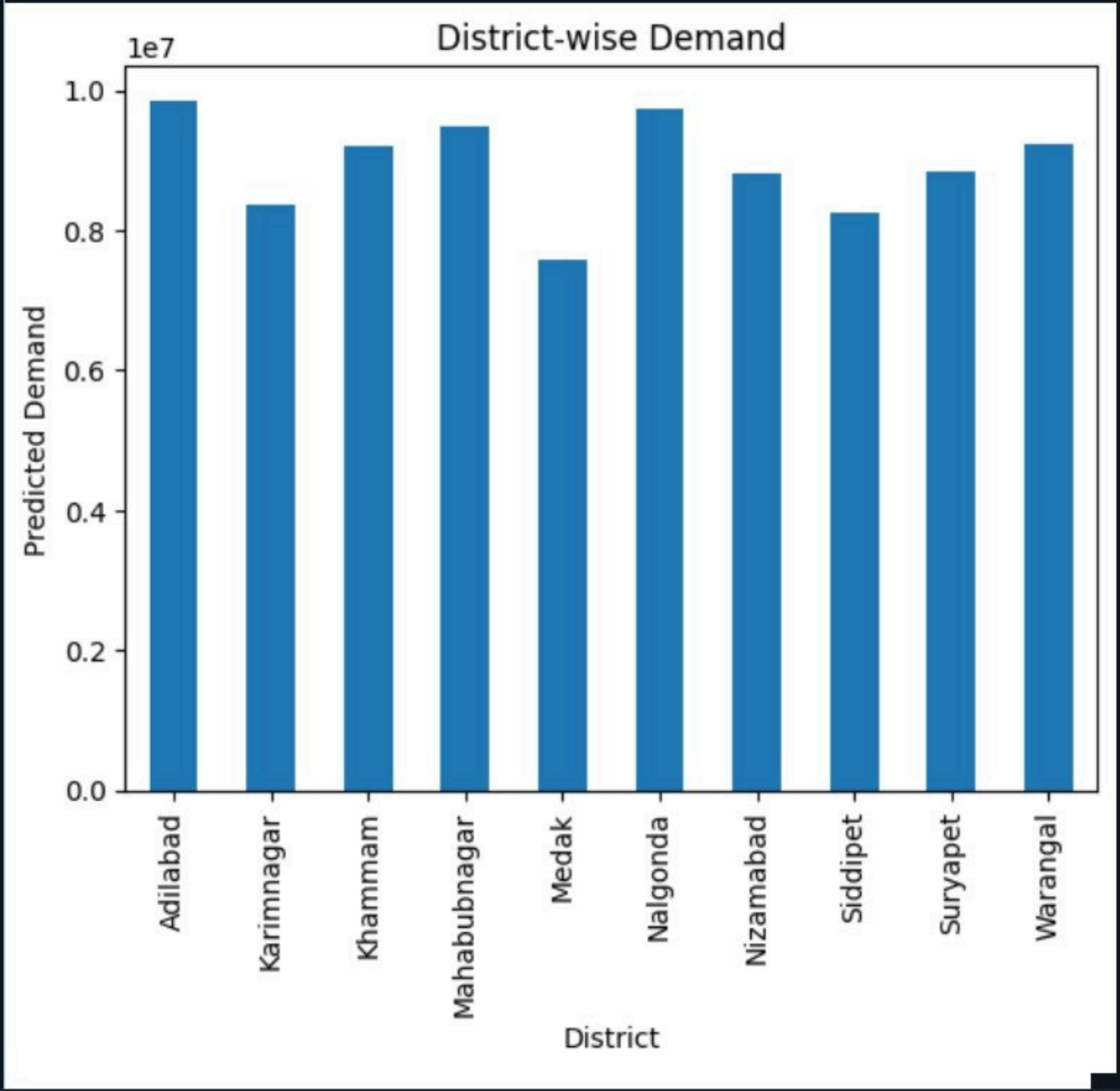
tration_analysis

File Edit View Run Help Python Last edit was 2 days ago Run all tration-cluster Schedule

District

Table

	uggestion	1.2 hunger_risk	1.2 allocation	ai_decision	crisis_alert	behavior_flag	anomaly_flag
2	105276	100	105276.150000000...	Increase Supply	true	false	false
3	108102	100	108102.75	Increase Supply	true	false	false
4	131570	100	131570.25	Stable	true	true	false
5	130701	100	130701.900000000...	Increase Supply	true	false	false
6	112143	100	112143.150000000...	Stable	true	false	false
7	126039	100	126039.900000000...	Stable	true	false	false
8	113555	100	113555.400000000...	Stable	true	false	false
9	138670	100	138670.35	Increase Supply	true	true	false
10	97644	100	97644.75	Increase Supply	true	false	false
11	80918	100	80918.25	Stable	true	true	true
12	43066	100	43066.8	Stable	true	true	false
13	133147	100	133147.35	Increase Supply	true	true	false
14	93433	100	93433.2	Increase Supply	true	true	false
15	47520	100	47520.9	Stable	true	false	false
16	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]


District-wise Demand
1e7
1.0
0.8
0.6
0.4
0.2
0.0
Predicted Demand
Adilabad
Karimnagar
Khammam
Mahabubnagar
Medak
Nalgonda
Nizamabad
Siddipet
Suryapet
Warangal
District

databricks

Search data, notebooks, recents, and more... ⌘ + P

T-Retlen

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- Workspace
- Data catalog
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- Experiments
- Experiments
- Ration Models
- Ration Models
- Experiment group
- Analytics
- Data analyses
- Deceriment group
- Workspace Setings

00:01 AM (7m) 6 Python

```python
ieport pandas as pd
from sklearn.model_selection ieport train_test_split
from sklearn.linear_madel iaport LinearRegrescion
from sktoarn.metrice ieport meen_squared_errer, r2_score

ration_data = pd.read_csv("/dbfs/FileStore/tables/ration_distributien_data_v2.csv")

X = ration_data[['tetal_cards', 'transaction_count', 'available_stock']]
y = ration_data["actual_demand']

X_train, X_tost, y_train, y_test = train_test_split(
    X, y, test_size=0.2, rendom_state=42
)

deeand_model = LinearRegression()
demand_model.fit(X_train, y_train)

y_pred = donand_model.predict(X_tect)
priet("Forecast Mean Iqoered Erron:", mean_squared_error(y_test, y_pred))
print("Forecast R^3 Scere:", r2_ssore(y_test, y_pred))
```

(1) MLflow run

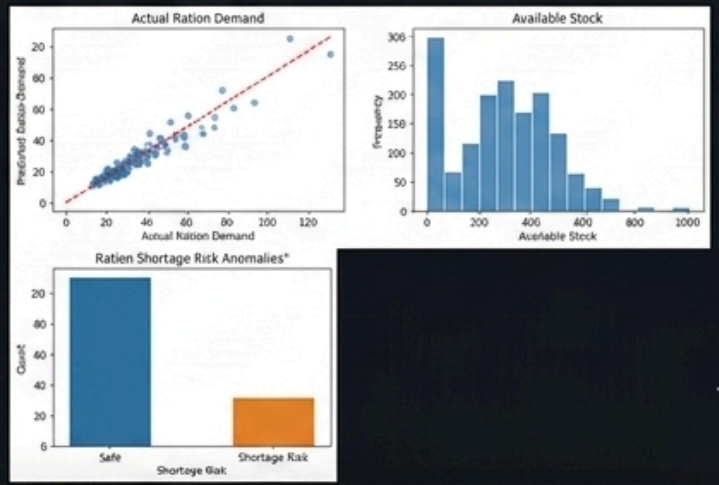
Legged 1 run for a forecasting experiment in MLflow.

```
> data: pandas.core.frame.DataFrame = (total_cords: int64, transection_count: int64, available_steck: int64, actual_demand: int64 ...)
> X: pandas.care.frame.DataFreme = (total_carde: int64, transaetien_ceonh: int64, available_stock: int64)
> X_tect: pandes.core.frame.DotaFrame = (bmount: int64, transastien_sounk: int65)
> 3_bain: pandas.core frame.DateFreme = (amout: int64, transaction_count: int64)

Forecast R^3 Scere: 6.94
Forecast Mean Squared Error: 15.2
```

00:05 AM (7m) Python

```python
ieport matpletlib.pyplet as plt
ieport seabern as sns
import * Added for better charts

# Define ration shertage threskeld (e.g., if predicted demand exceeds steck)
ration_data['shortage_risk'] = (y_pred > X_test['available_stosk'])

# Plot actual vs predicted deeand
plt.figure(figsise=(10,9))
plt.ssatter(y_test, y_pred, alpha=0.3)
plt.plet([ionnty_test], max(y_test)], [min(y_test), max(y_test)], '--r')
plt.pletf[ean(?_?e?t), max(?_1411)], [etnty_test), naxty_test)], '--r')
plt.xlabel("Actual Ration Demand (kgs)")
plt.ylabel("Predicteä Natian Deeand (kgs)")
plt.title("Ration Demand: Actual vs Predicted")
plt.shew()

# Plot distribution of stock availability
plt.figure(figsize=(10,3))
plt.hist(ration_date['available_stock'], bins=20, alpha=0.7)
plt.xlakell('Available Steck (kgs)')
plt.title("?-Ratien Snventery Tracking")
plt.shew()

# Plot aoeealy distribution
plt.figure(figsize=(6,4))
anemaly_counts = ration_dota('skortage_risk').value_counts()
anemaly_coonts.ptat(kine='ser')
plt.xtloks(tiokn=19, 1], leeels=['Safe', 'Shortage Risk'])
plt.ylabel("Caunt")
plt.title("Ration Shortage Risk Anomalies")
plt.show()
```

00:05 AM (7m) Python Python

```python
inport matpletlib.pyplot as plt
import seaborn as sns

# Define ration shortage threshold (e.g., if predicted demand exceeds steck)
plt.figure(figsize=(10,F))
ptt.scatter(y_test, y_pred, alpha=0.5)
plt.plot([ninly_test), meviy_tezt)), [min(y_test), max(y_test)], '--r')
plt.xlabel("Actsasl Ration Sanand (kgs)")
ptt.ylabel("Predicted Ratien Deeeed (kgs)")
ptt.title("Ration Demand: Actual vs Predicted")
plt.show()

# Plot distribution of stock availability
ptt.xlabel("Available Stock (kgs)")
pts.slabel("Frequency")
ptt.title("T-Rekien Inventery Tracking")

# Plot anoealy distributien
plt.figure(figsize=(5,4))
anoealy_counts = retten_data['shortage_risk'].value_counts()
anoealy_sounts.plat(kind="bar")
pl8.xtisks(ticks=[0, 1], labels=['Safe', 'Shortege Risk'])
plt.ylabel("Count")
ptt.show()
```

Anomaly detection identified 15 regions with potential shortage risk.